

EXPERIMENTAL REPORT

Multistage Amplifier Design, Simulation and Measurement

Multistage Amplifier

Two-stage 2N7002 amplifier using SPICE simulation and hardware verification

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Report scope

This report follows the provided experimental instruction manual: establish a two-stage amplifier in SPICE, measure the DC operating point, run transient analysis, run small-signal AC analysis, and verify input/output resistance and frequency response using simulation and oscilloscope screenshots.

1. Objective

The purpose of this experiment is to design a two-stage multistage amplifier with SPICE, verify its bias point and small-signal behavior, and compare the simulated results with measurements taken from the physical circuit.

The main performance items are voltage amplification, frequency response, input resistance, and output resistance.

Expected learning outcomes

- Build a two-stage MOSFET amplifier using 2N7002 devices, bias resistors, source/drain resistors, coupling capacitors, and a load resistor.
- Use DC operating point analysis to confirm that both transistor stages are properly biased.
- Use transient and AC analysis to evaluate gain, phase, passband behavior, and cutoff trends.
- Use voltage-divider and loaded/unloaded-output methods to estimate input and output resistance.

Data note

The report uses numerical values that are readable in the provided screenshots. When a screenshot does not show exact cursor readings, the result is described qualitatively instead of inventing unsupported numbers.

2. Circuit Design and Experimental Method

The amplifier consists of two cascaded common-source MOSFET stages. Coupling capacitors block DC between the source, amplifier stages, and load, while the resistor networks establish the DC gate bias. The load resistance is 10 k Ω and the supply voltage is 12 V.

Item	Value / Model	Function
M1, M2	2N7002	Active amplifying devices
VDD	12 V	DC supply
Input signal	SINE(0, 10 mV, 1 kHz); AC = 1	Transient and AC excitation
C1, C2	1 μ F	Input and output coupling
R1	1 k Ω	Series/source resistance in input path
R2 / R5	118 k Ω / 240 k Ω	Gate-bias divider for the first stage
R3	1.8 k Ω	First-stage source resistor
R6	7.3 k Ω	First-stage drain resistor / second-stage gate bias node
R4	1 k Ω	Second-stage source resistor
R7	8.5 k Ω	Second-stage drain resistor
RL	10 k Ω	Output load

Procedure summary

- Establish the two-stage amplifier circuit in LTspice.
- Run DC operating point analysis to observe all node voltages and device currents.
- Apply a sinusoidal input signal and run transient analysis to compare input and output waveforms.
- Set a small-signal AC source and run AC analysis to obtain the magnitude and phase response.
- Measure input resistance using a known series resistor and compare source voltage with the input-port voltage.
- Measure output resistance using the no-load output voltage and the loaded output voltage.

3. Simulation Setup and DC Operating Point

The LTspice circuit screenshot shows the complete two-stage amplifier. The input source is AC-coupled through C1, the two 2N7002 stages are cascaded, and the output is coupled through C2 to the 10 k Ω load.

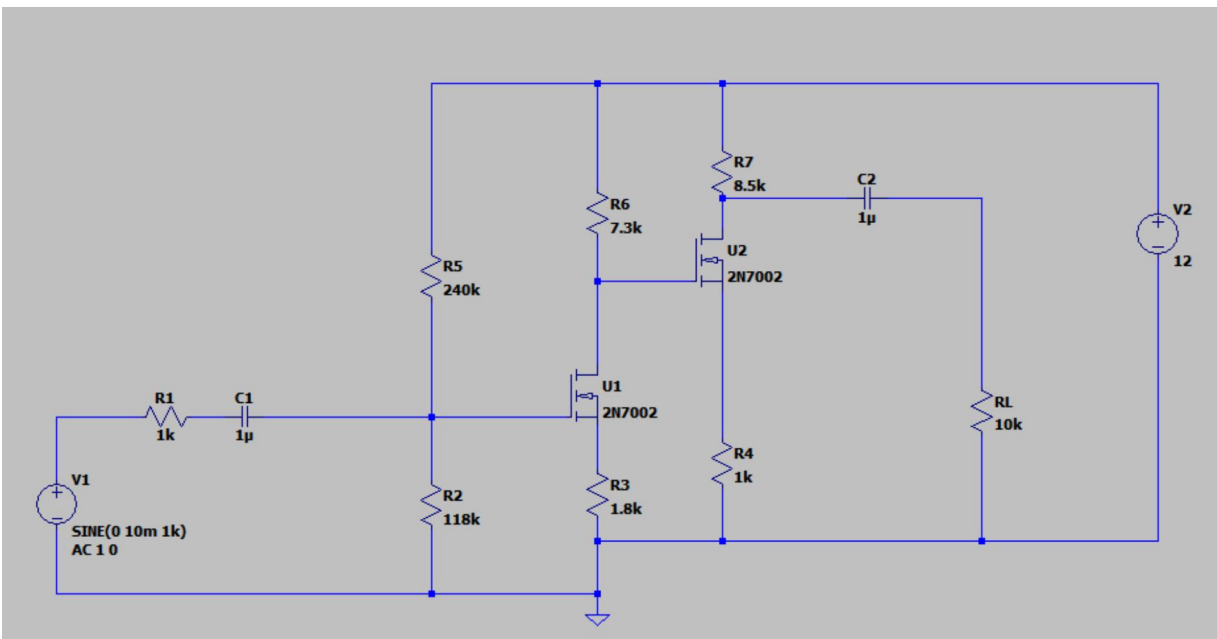


Figure 1. LTspice schematic of the two-stage 2N7002 multistage amplifier.

The DC operating point confirms that both MOSFET stages conduct approximately 1.2 mA. This indicates that the bias network places the devices in an operating region suitable for small-signal amplification.

Quantity	Value from screenshot	Comment
V(n001)	12 V	Supply rail
V(n002)	1.62204 V	First-stage gate-bias node
V(n004)	2.9417 V	Intermediate drain/gate coupling node
V(n005)	1.22094 V	First-stage source node
V(n007)	3.95531 V	Second-stage drain/output-side node
Id(M1)	1.22094 mA	First-stage drain current
Id(M2)	1.24086 mA	Second-stage drain current

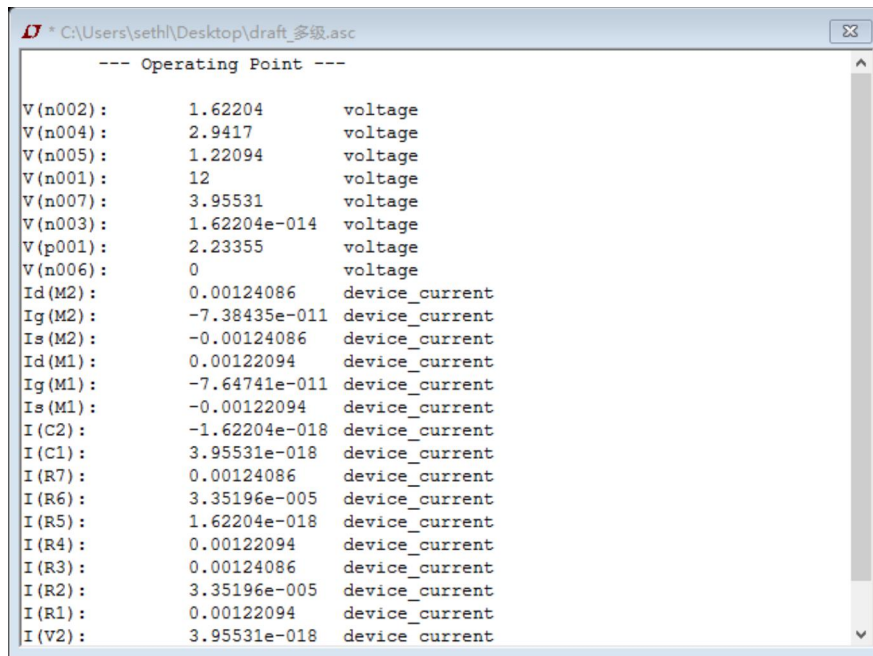


Figure 2. LTspice DC operating point results.

4. Transient and AC Small-Signal Analysis

Transient analysis at 1 kHz shows sinusoidal behavior at the input and output. The output waveform is larger than the input waveform and remains smooth, so the amplifier is operating without obvious clipping under the applied 10 mV input condition.

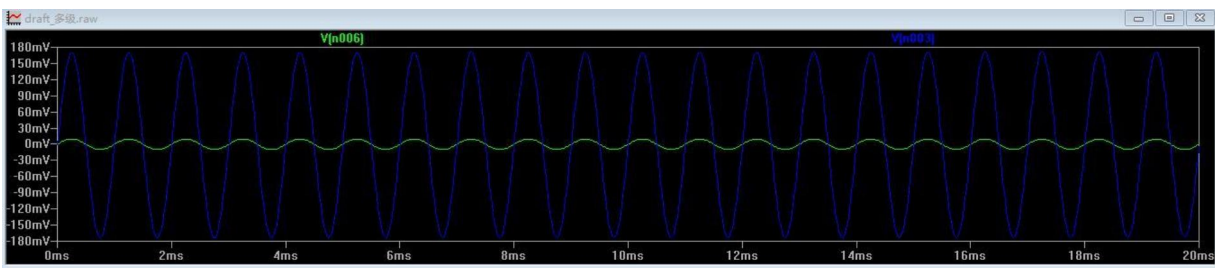
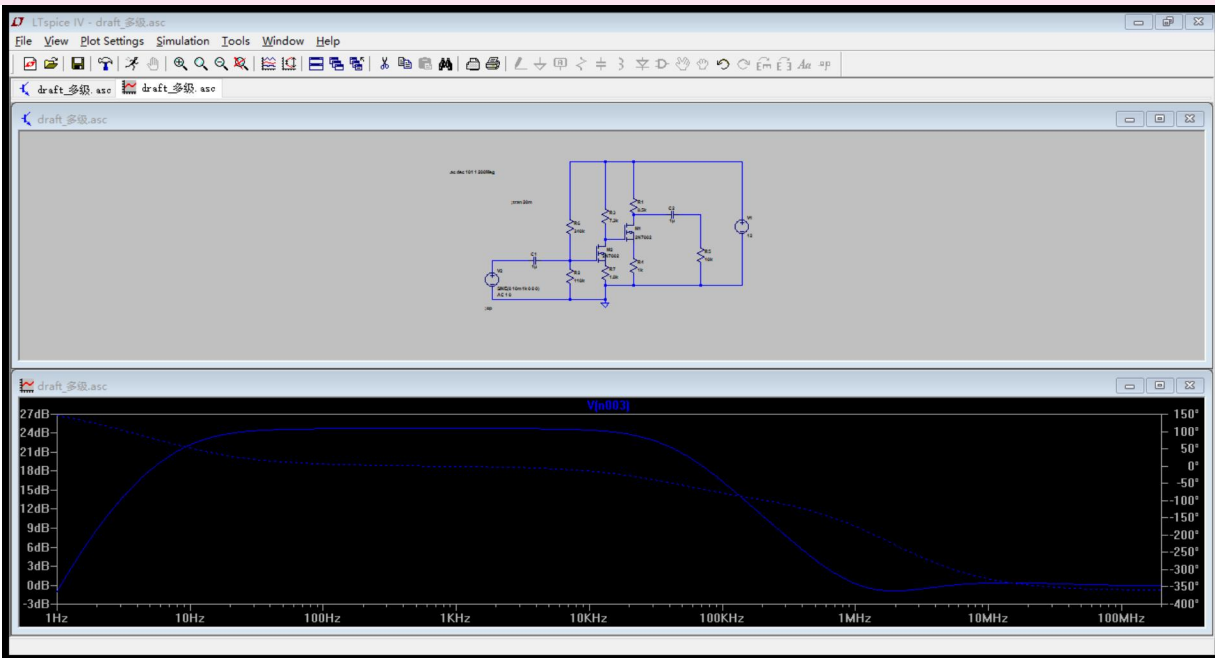


Figure 3. Simulated transient response at 1 kHz.

The AC analysis result shows a midband gain near the flat part of the magnitude curve, approximately 24 dB in the screenshot. The low-frequency roll-off is mainly caused by the coupling capacitors, while the high-frequency roll-off is caused by device capacitances and the finite bandwidth of the cascaded stages. The dotted curve indicates phase variation with frequency.



5. Input and Output Resistance Evaluation

The input resistance is measured by inserting a known resistor R_s between the source and the amplifier input. The output resistance is measured by comparing the no-load output voltage with the loaded output voltage.

Parameter	Measurement relation	Meaning
Input resistance	$R_{in} = V_i / (V_s - V_i) \times R_s$	V_i is the amplifier input-port voltage and V_s is the source voltage.
Output resistance	$R_{out} = (V_o / V_L - 1) \times R_L$	V_o is no-load output voltage and V_L is load voltage.

In the input-resistance simulation, the source and input-port voltages are plotted together. The smaller voltage drop across the inserted resistor corresponds to a larger input resistance.

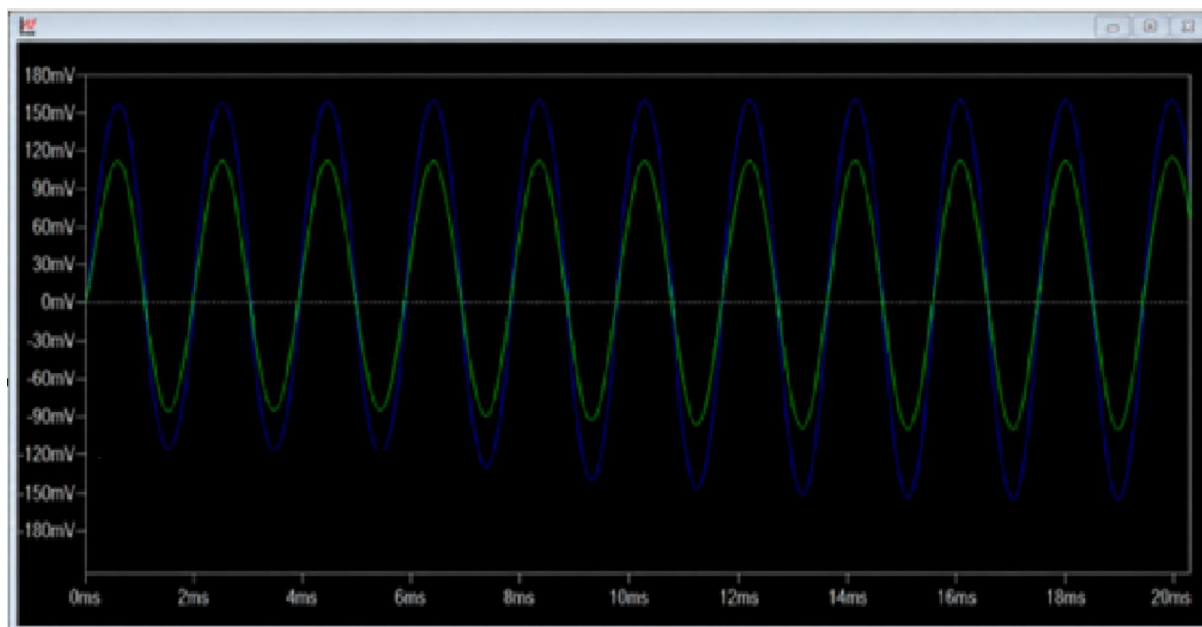


Figure 5. Input-resistance simulation waveform.

In the output-resistance simulation, the loaded and no-load output voltages are compared. A larger difference between the two waveforms indicates a larger output resistance relative to the external load.

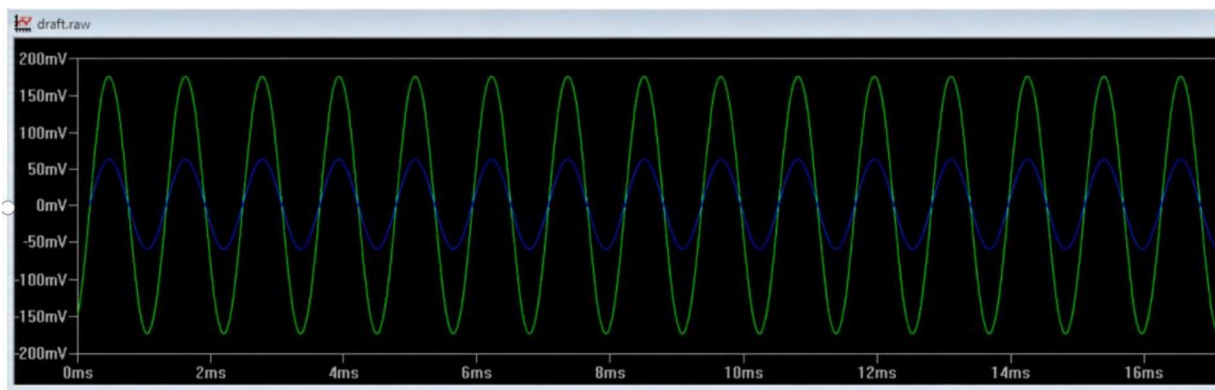


Figure 6. Output-resistance simulation waveform.

6. Hardware Implementation

The physical circuit was assembled on a breadboard. The photo shows the MOSFETs, resistors, capacitors, jumper wires, and measurement connections. Breadboard wiring and component tolerances introduce parasitic resistance and capacitance, so the measured response may differ from the SPICE model, especially at high frequency.

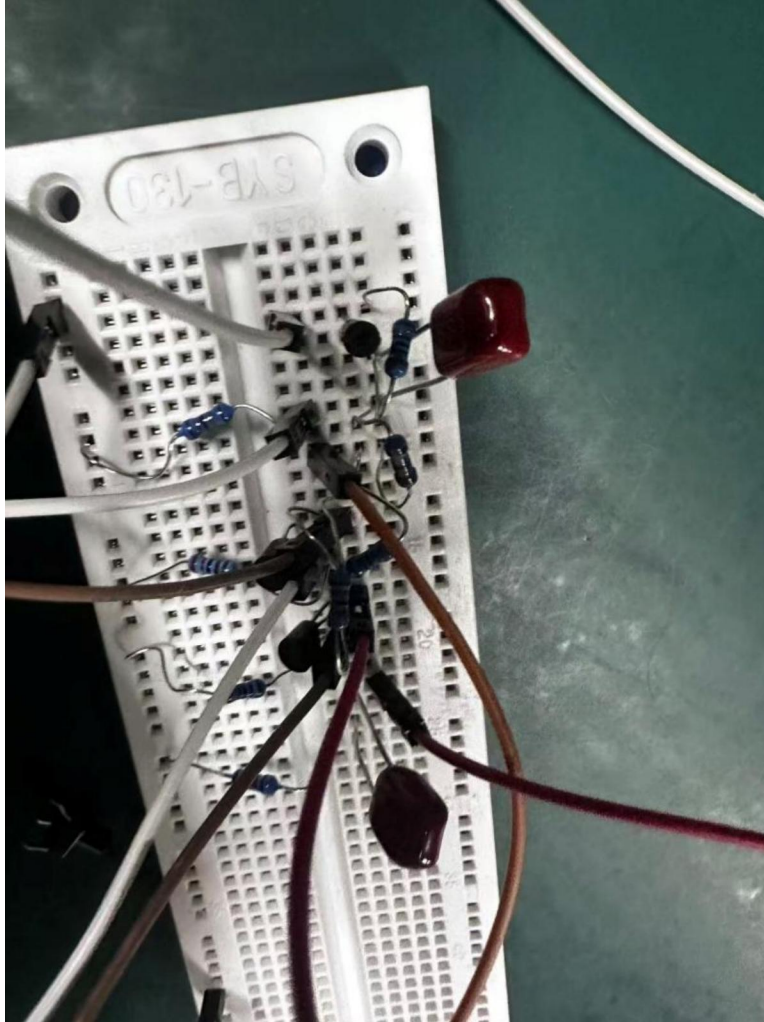


Figure 7. Breadboard implementation of the multistage amplifier.

7. Oscilloscope Frequency-Response Measurements

The oscilloscope screenshots were taken at 1 kHz, 10 kHz, and 100 kHz. The measured waveforms are consistent with the simulated frequency-response trend: the midband sine wave is stable, while the high-frequency output is visibly attenuated.

Frequency	Visible observation	Interpretation
1 kHz	Stable sinusoidal output waveform	Passband operation
10 kHz	$V_{pp} \approx 296\text{ mV}$ and $V_{rms} \approx 103\text{ mV}$ are displayed on the scope	Still in useful midband region
100 kHz	Output amplitude is visibly smaller than at 10 kHz	High-frequency roll-off becomes significant

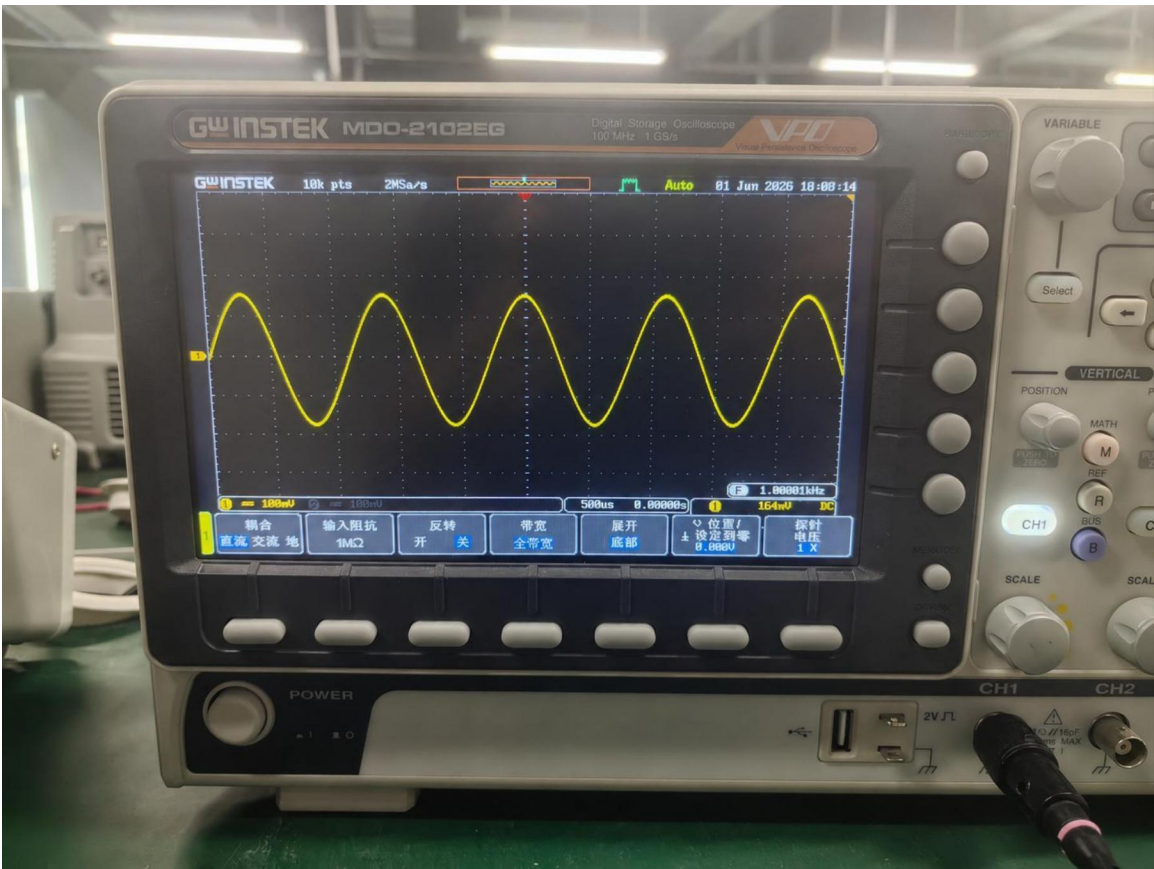


Figure 8. Measured output waveform at 1 kHz.

7. Oscilloscope Frequency-Response Measurements (continued)

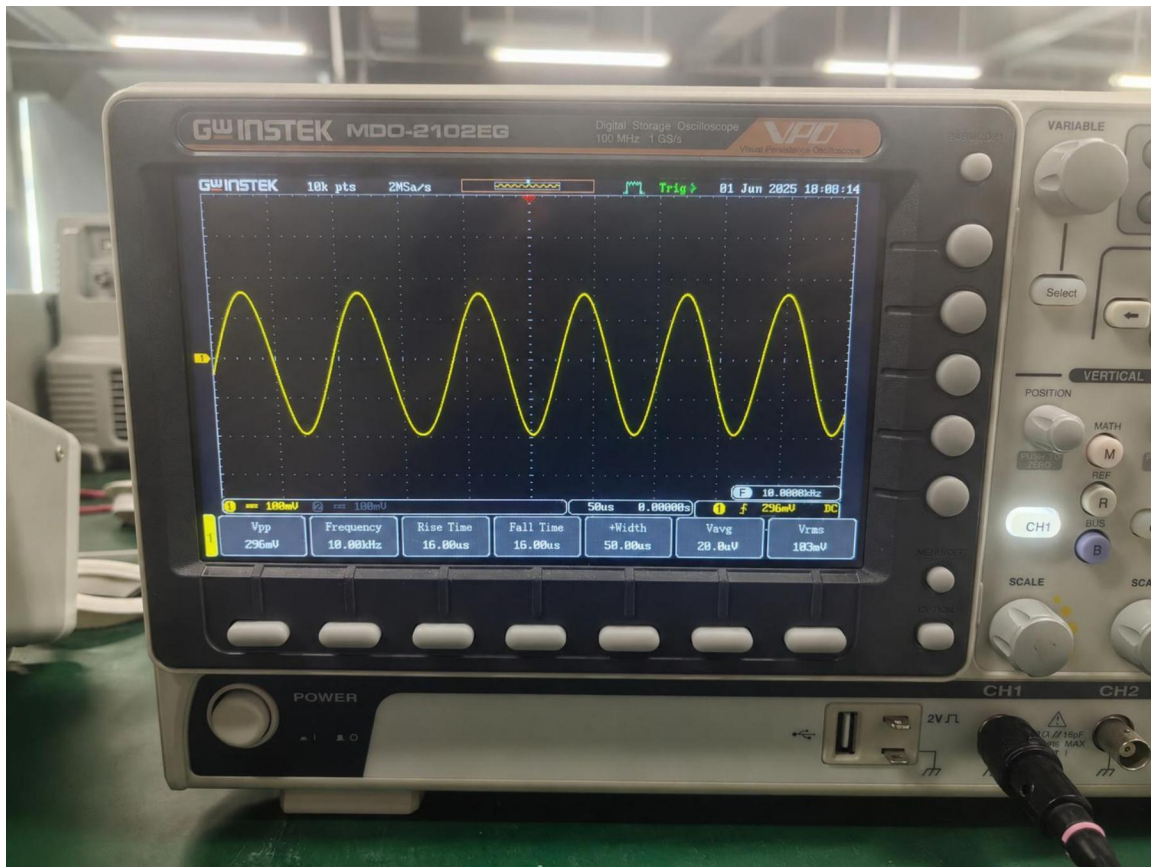


Figure 9. Measured output waveform at 10 kHz.

At 10 kHz, the oscilloscope measurement panel shows approximately 296 mV peak-to-peak and 103 mV RMS. The waveform remains sinusoidal, which agrees with the flat midband region in the AC simulation.

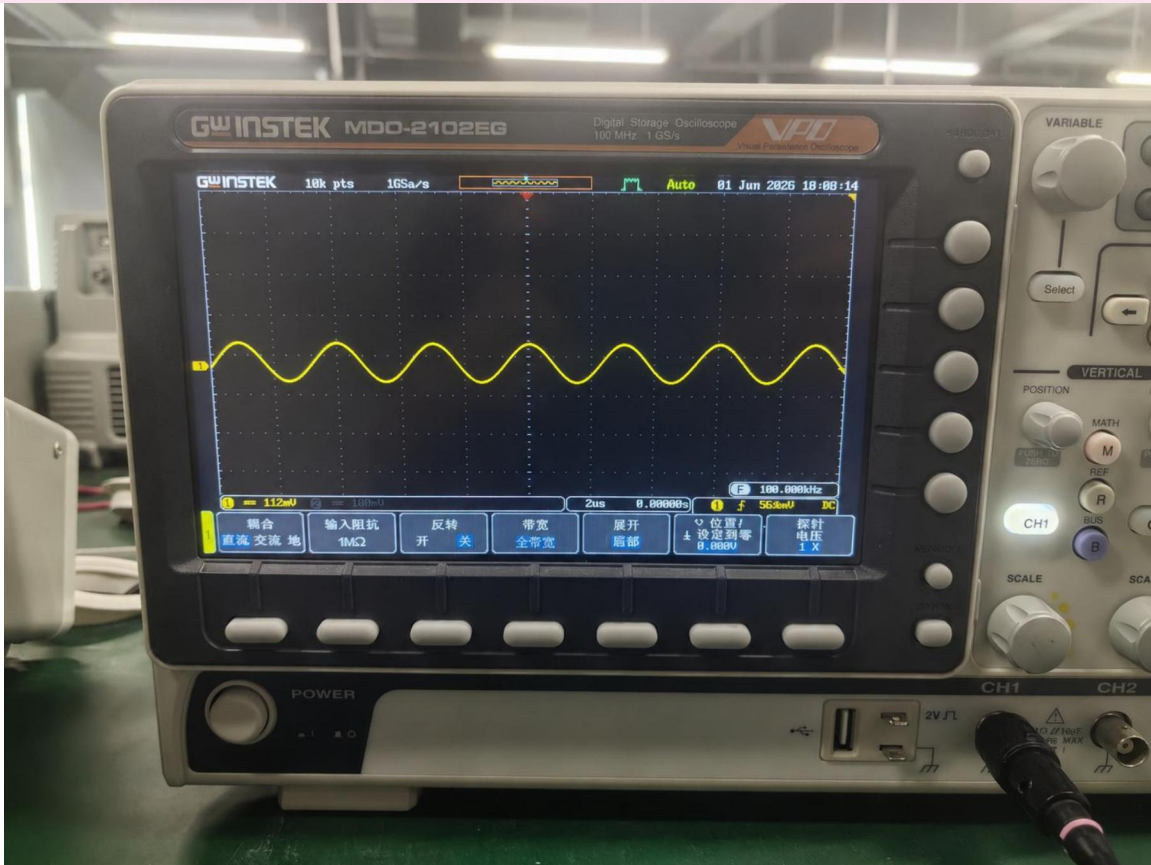


Figure 10. Measured output waveform at 100 kHz.

At 100 kHz, the output is still sinusoidal but its amplitude is much smaller. This result agrees with the simulated high-frequency gain reduction.

8. Resistance Measurement Results

The input-resistance measurement compares two 1 kHz waveforms. The oscilloscope screen shows approximately 19.9 mV peak-to-peak for one channel and 19.8 mV peak-to-peak for the other, so the voltage drop across the inserted series resistor is very small. This indicates that the amplifier input resistance is much larger than the inserted resistor R_s .

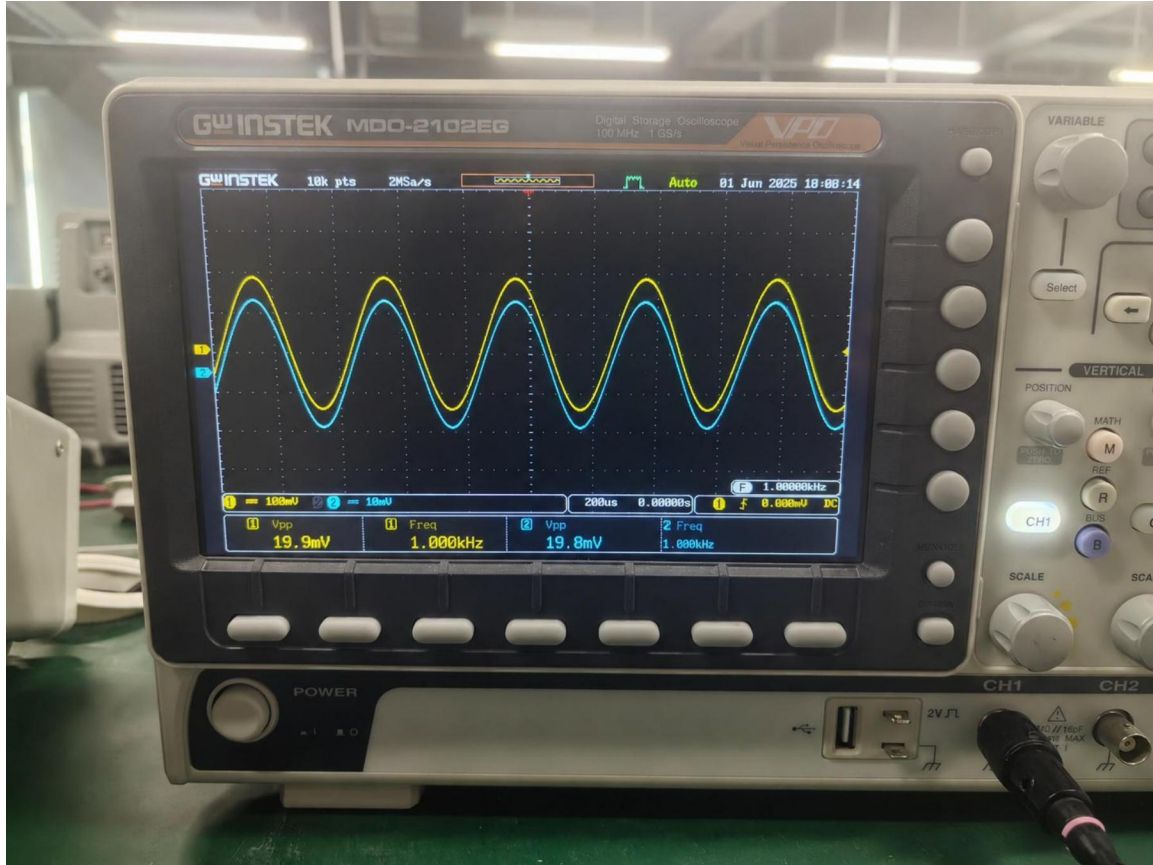


Figure 11. Input-resistance measurement waveform.

8. Resistance Measurement Results (continued)

For output resistance, the loaded and no-load output voltages are compared. The screenshot displays approximately 576 mV peak-to-peak for one channel and 320 mV peak-to-peak for the other channel. If the 576 mV waveform is taken as V_o and the 320 mV waveform as V_L with $R_L = 10\text{ k}\Omega$, then $R_{out} \approx (576/320 - 1) \times 10\text{ k}\Omega \approx 8.0\text{ k}\Omega$.

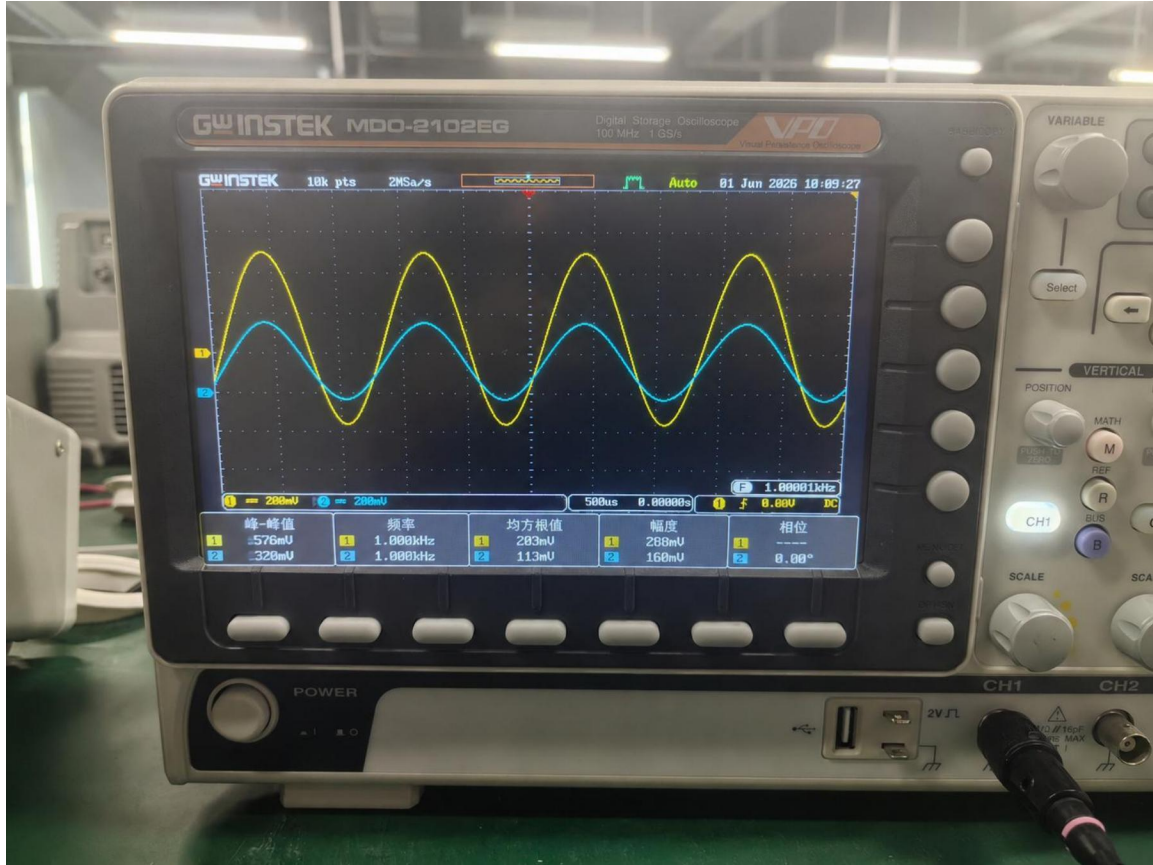


Figure 12. Output-resistance measurement waveform.

9. Discussion

The simulation and measurement results show the same general behavior. The amplifier gives clear voltage amplification in the midband, and the waveform quality is good at 1 kHz and 10 kHz. The smaller amplitude observed at 100 kHz is consistent with the upper cutoff trend shown by the AC simulation.

The DC operating point is reasonable because both 2N7002 stages have drain currents close to 1.2 mA. The output resistance estimated from the measured waveforms is also physically reasonable because it is close to the second-stage drain resistance of 8.5 k Ω .

Main error sources

- MOSFET model differences: the actual 2N7002 parameters may not exactly match the LTspice model.
- Breadboard parasitics: long jumper wires and breadboard capacitance reduce high-frequency performance.
- Instrument loading: oscilloscope probes and generator output impedance affect measured voltages.
- Component tolerances: resistor and capacitor tolerances shift the bias point and cutoff frequencies.
- Reading resolution: screenshots without cursor data limit exact numerical extraction.

10. Conclusion

A two-stage 2N7002 multistage amplifier was designed, simulated, built, and measured. The LTspice operating point confirms stable biasing, the transient simulation confirms sinusoidal amplification, and the AC analysis shows a clear passband with low- and high-frequency roll-off. Hardware measurements at 1 kHz and 10 kHz show stable sinusoidal output, while the 100 kHz waveform is attenuated, matching the expected high-frequency roll-off. The input-resistance measurement suggests a high input resistance, and the output-resistance measurement gives an estimated R_{out} of about 8.0 k Ω under the stated channel assignment.

Final result

The experiment meets the objective of using SPICE and physical measurements to evaluate the gain, frequency response, input resistance, and output resistance of a multistage amplifier.

Source Materials Used

The report was prepared from the provided experimental instruction manual and the provided simulation/measurement screenshot document. All figures in this report come from the screenshot document.